

Business Requirements for Pre-prepared Dishes Ordering System

By CS5200 Fall 2024 semester Qiong Wu Project 1

Inspired by Locale <https://www.ycombinator.com/launches/lgH-locale-we-re-making-restaurant-delivery-affordable>

Highlight **nouns** and **verbs**

Introduction

Compared to the flexibility and diversity of China's **food delivery industry**, the Bay Area's system feels limited and cumbersome. Inspired by **Locale's model** and the **experience of sharing Uber rides**, the concept of **consolidating multiple orders** into a **single delivery trip** **offers** a tangible solution. **Grouping pre-prepared dishes** from different **suppliers** into one **optimized delivery route** **reduces** **operational expenses**, **fuel costs**, and provides an **eco-friendlier system**, while offering flexibility to **customers** and **easing** the pressure on **local restaurants** during **peak hours**.

Problem Statement

Local restaurants **face** challenges from surging **demand** during peak hours and **quality issues** **caused** by **long-distance**, **real-time delivery**. These factors **increase** **pressure** on suppliers and lead to **higher costs** for both customers and restaurants. Current **platforms** **focus on** real-time deliveries, requiring **multiple orders** from nearby restaurants, which **inflates delivery fees** and **complicates** the **process**. Meanwhile, each **meal** is typically ordered from one restaurant, **resulting** in a high **proportion** of the cost being **attributed** to delivery fees. If customers want to **order** from another nearby restaurant, they are required to **place** a separate order, increasing costs further.

My Solution

My **pre-prepared dishes ordering system** will **focus on** meals from local suppliers, consolidating **orders** and offering flexible **delivery windows** to **optimize** restaurant **workloads**. By **avoiding**

peak traffic hours and using the best possible delivery routes, the system ensures an enhanced experience for drivers and restaurants. The key features of the database design include:

1. **Flexible Meal Bundles:** Customers can create their own meal bundles from various local restaurants.
2. **Flexible Delivery Windows:** Customers select flexible delivery windows, allowing restaurants to prepare meals without the rush of real-time orders.
3. **Optimized Route Delivery:** The system consolidates orders from multiple suppliers into a single delivery trip, using Google Maps API and OR-Tools, reducing fuel and time costs for drivers. (idea from ChatGPT)
4. **Cost Reduction for Restaurants:** By spreading out delivery and allowing restaurants to prepare meals at less congested times, overhead costs are lowered, leading to better margins.

Conclusion

By combining ideas from Locale's innovative model and the ride-sharing concept, this system creates a streamlined, cost-effective, and environmentally friendly meal delivery service. It optimizes logistics, enhances supplier efficiency, and improves the overall customer experience, offering a competitive advantage over traditional food delivery platforms.

Collect the nouns and verbs

Nouns

- food delivery industry
- the Bay Area
- Locale's model
- sharing Uber rides
- multiple orders
- orders
- single delivery trip
- pre-prepared dishes
- suppliers
- optimized delivery route
- operational expenses
- fuel costs

- eco-friendlier system
- customers
- local restaurants
- peak hours
- demand
- quality issues
- long-distance, real-time delivery
- pressure
- higher costs
- platforms
- delivery fees
- proportion
- pre-prepared dishes ordering system
- delivery windows
- workloads
- experience
- drivers
- Meal Bundles
- Google Maps API
- OR-Tools
- fuel
- time costs
- logistics
- supplier efficiency
- advantage

Verbs

- consolidating
- offers
- Grouping
- reduces
- easing
- face
- caused
- increase
- focus on
- inflates
- complicates
- resulting
- attributed
- optimize

- avoiding
- using
- ensures
- create
- select
- prepare
- spreading
- combining

Reorganize the Nouns and Verbs. Classify them

Nouns

- food delivery industry
 - the Bay Area
- orders
 - optimized delivery route
 - operational expenses
 - fuel costs
 - long-distance, real-time delivery
- suppliers
 - local restaurants
 - supplier efficiency
- customers
 - demand
 - orders
- delivery
 - fees
 - proportion
- pre-prepared dishes ordering system
 - single delivery trip
 - pre-prepared dishes
 - delivery windows
 - Meal Bundles
 - Google Maps API
 - OR-Tools
- drivers
 - fuel
 - time costs
 - logistics

Verbs

- create

- offers
- grouping
- optimize
- avoiding
- select
- prepare
- spreading
- combining

Rules of Business

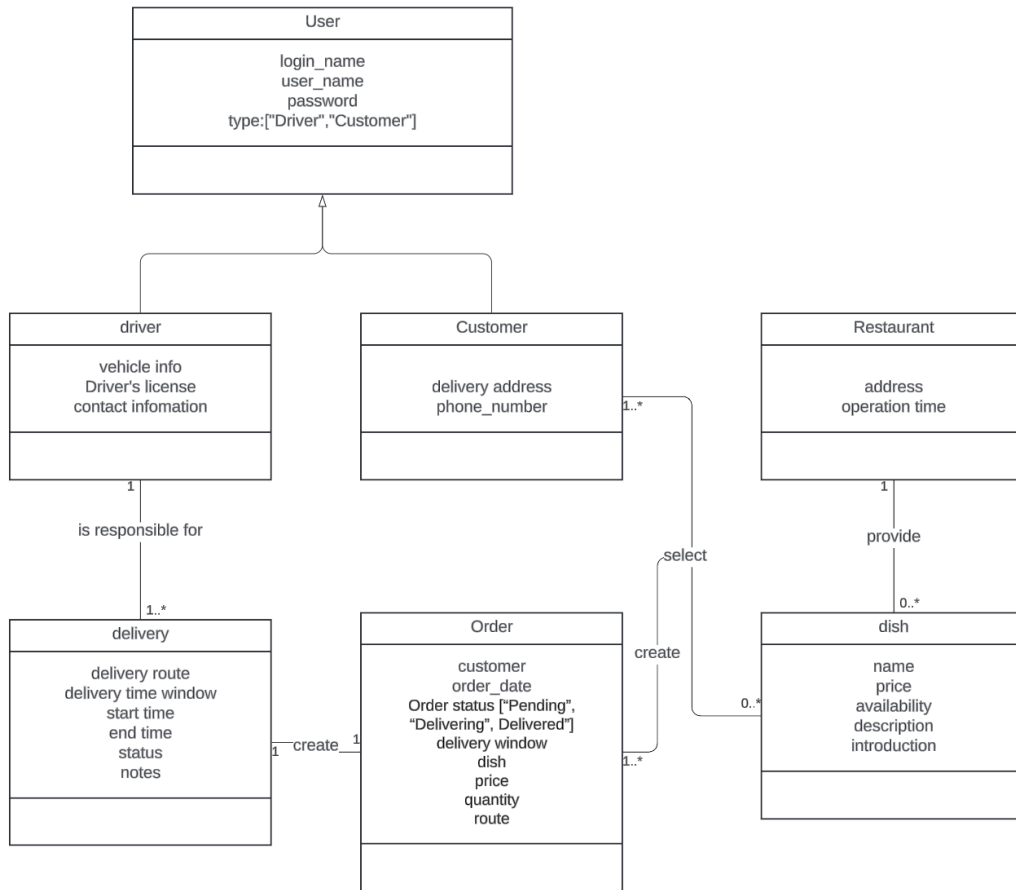
- Any user can create accounts as customer, but they must provide valid address and phone number.
- The accounts for supply restaurants will be created after approval following a partnership agreement.
- Individuals can create delivery accounts, but they must provide a valid driver's license and vehicle information.
- Each supplier can only provide one menu but can have multiple dishes with price and availability quantity.
- Customers can place an order consisting of items from multiple suppliers.
- All selected items from different suppliers are combined into a single order for delivery.
- The system will analyze traffic patterns and supplier preparation times to suggest optimal delivery windows.
- Drivers can select orders, and once the driver selects the orders, a suggested pickup sequence is created.
- Suppliers can update their menu offerings, including prices, availability, and descriptions. But the price of existing orders will not change.

Extract classes

- Customer
 - Login
 - Password
 - Name
 - Address
 - Phone number
- Suppliers
 - Name
 - Address
 - Opening hours

- Menu
- Supplier Menu
 - Supplier
 - Price
 - Availability
 - Description
 - Introduction
- Order
 - Customer
 - Order date
 - Order status [“Pending”, “Delivering”, Delivered”]
- Order items
 - Item
 - Price
 - Quantity
- Delivery
 - Order
 - Driver
 - Pickup Time
 - Delivery Time
- Delivery route
 - Start location
 - End location
 - Stops
 - Sequence
- Delivery Windows
 - Date
 - Start time
 - End time
- Driver
 - Name
 - Vehicle Info
 - Driver’s license
 - Phone Number

UML Class Diagram



ERD

